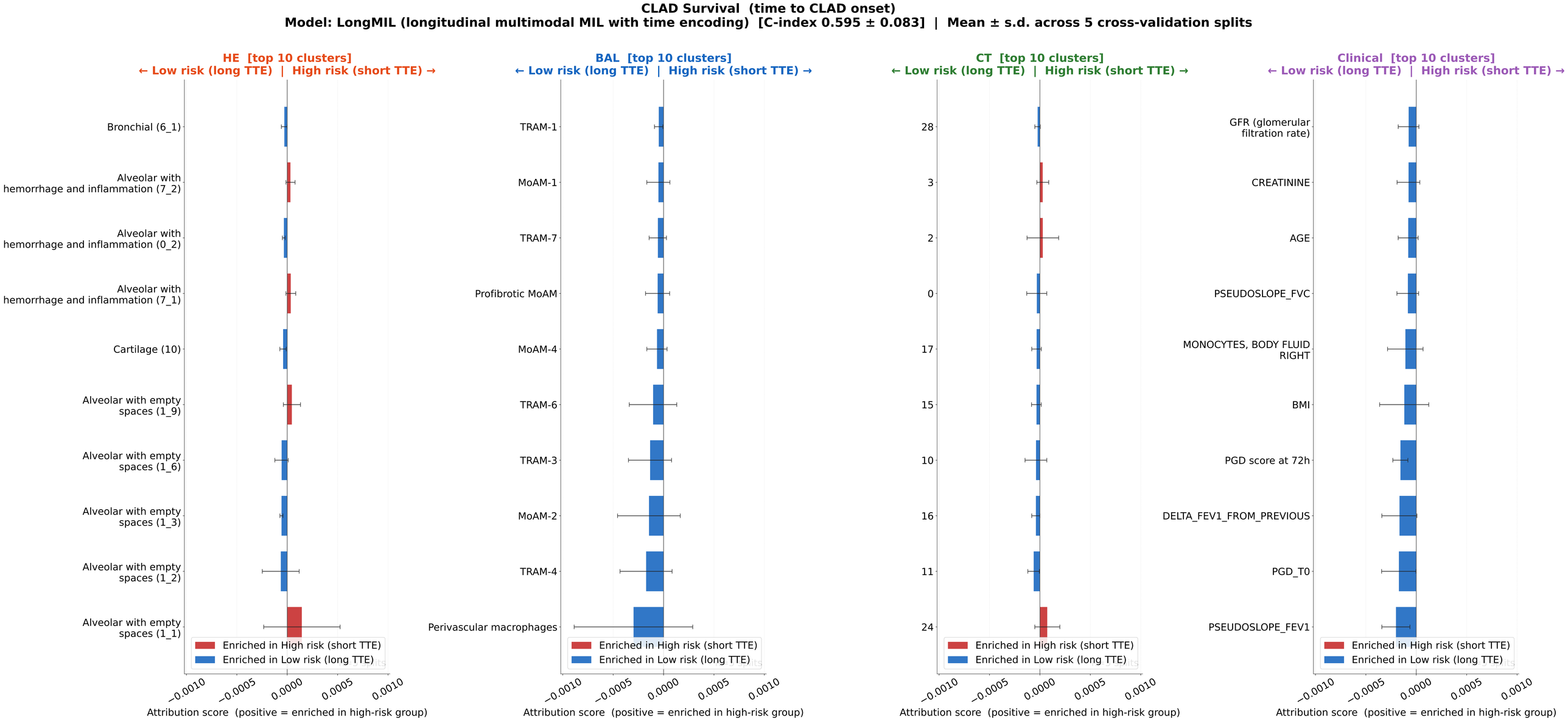
CLAD Survival (time to CLAD onset)
Model: LongMIL (longitudinal multimodal MIL with time encoding) [C-index 0.595 ± 0.083] | Mean ± s.d. across 5 cross-validation splits
HE [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
Bronchial (6_1)
Alveolar with hemorrhage and inflammation (7_2)
Alveolar with hemorrhage and inflammation (0_2)
Alveolar with hemorrhage and inflammation (7_1)
Cartilage (10)
Alveolar with empty spaces (1_9)
Alveolar with empty spaces (1_6)
Alveolar with empty spaces (1_3)
Alveolar with empty spaces (1_2)
Alveolar with empty spaces (1_1)
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
BAL [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
TRAM-1
MoAM-1
TRAM-7
Profibrotic MoAM
MoAM-4
TRAM-6
TRAM-3
MoAM-2
TRAM-4
Perivascular macrophages
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
CT [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
28
3
2
0
17
15
10
16
11
24
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)
Clinical [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
GFR (glomerular filtration rate)
CREATININE
AGE
PSEUDOSLOPE_FVC
MONOCYTES, BODY FLUID RIGHT
BMI
PGD score at 72h
DELTA_FEV1_FROM_PREVIOUS
PGD_T0
PSEUDOSLOPE_FEV1
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0010
−0.0005
0.0000
0.0005
0.0010
Attribution score (positive = enriched in high-risk group)